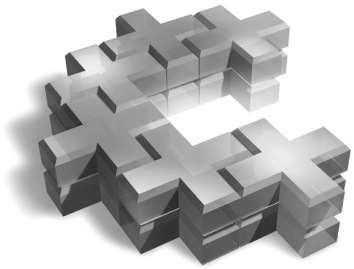
Microsoft® Official Curriculum



Mastering MFC Development

Student Workbook

Companion to the Mastering MFC Development Using Microsoft® Visual C++® 6 Compact Disc

Course Number: 1015A

Part Number: X03-93113
Release Date: July 1998

Microsoft®

Information in this document is subject to change without notice. The example companies, organizations, products, people, and events depicted herein are fictitious. No association with any real company, organization, product, person or event is intended or should be inferred. Complying with all applicable copyright laws is the responsibility of the user. Without limiting the rights under copyright, no part of this document may be reproduced, stored in or introduced into a retrieval system, or transmitted in any form or by any means (electronic, mechanical, photocopying, recording, or otherwise), or for any purpose, without the express written permission of Microsoft Corporation.

Microsoft may have patents, patent applications, trademarked, copyrights, or other intellectual property rights covering subject matter in this document. Except as expressly provided in any written license agreement from Microsoft, the furnishing of this document does not give you any license to these patents, trademarks, copyrights, or other intellectual property.

©1998 Microsoft Corporation. All rights reserved.
Printed in the United States of America.

Microsoft, ActiveX, Developer Studio, FoxPro, FrontPage, IntelliSense, SourceSafe, Visual Basic, Visual C++, Win32, Windows, and Windows NT are registered trademarks and MSDN, Visual SourceSafe, and Visual Studio are trademarks of Microsoft Corporation in the United States of America and other countries.

Other product and company names herein may be the trademark of their respective owners.

Course Number: 1015A
Part Number: X03-93113

Contents

Chapter 1: Introduction to Visual C++ 6.0 1
Overview of Visual C++ and MFC 2
Features of Visual C++ 3
Features of the MFC Library 5
Creating an MFC-based Application 7
Enhancements to Visual C++ 8
Debugger 9
Editors 10
Projects 11
Wizards 12
Tools 13
Enhancements to MFC 15
Database 16
Internet 17
Chapter 2: Debugging and Error Handling 19
Debugging 20
Overview of the Visual Studio Debugger 21
Using the Edit and Continue Feature 22
Handling Errors and Exceptions 23
Writing Robust Functions 24
Handling Exceptions 26
Exception-Handling Process 28
MFC Exception Classes 30
Self-Check Questions 32
Chapter 3: Enhancing the User Interface 33
Enhancing Menus 34
Implementing Dynamic Menus 35
Implementing Cascading Menus 40
Implementing Ownerdraw Menus 43
Enhancing Toolbars 46
Enabling Docking in a Frame Window 47
Creating a Floating Toolbar 48
Enhancing Status Bars 51
Enhancing Dialog Boxes 57
Customizing Common Dialog Boxes 58
Extending Dialog Data Validation 60
Coordinating Controls 61
Creating Property Sheets and Tabbed Dialog Boxes 62
Using Modeless Dialog Boxes 65
Initializing and Displaying a Modeless Dialog Box 66
Implementing Command Buttons 68
Deleting the Dialog Box Object 69
Communicating with the Application 70
Using Dialog Bars 74
Introduction to Dialog Bars 75
Adding a Dialog Bar 76
Implementing Command Handlers 78
Using Rebars 80
Introduction to Rebars 81
Creating a Rebar 82

Adding Child Windows to a Rebar 84
Customizing Rebars 85
Sample Applications 86
Self-Check Questions 87
Chapter 4: Implementing View Classes 89
Introduction to Views 90
The CView Class 92
Managing Document/View Interaction 93
Adding Multiple Views 96
Adding Multiple Views to an SDI Application 97
Adding Multiple Views to an MDI Application 102
Suppressing Automatic Creation of a Child Window 110
Setting Titles for Child Frame Windows 111
Adding Scrolling Views 113
Introduction to Scrolling Views 114
Converting the Application View to a Scrolling View 115
Adjusting Scrolling Characteristics 116
Converting Coordinates 117
Implementing Splitter Windows 119
Introduction to Splitter Windows 120
Creating Dynamic Splitter Windows 121
Creating Static Splitter Windows 123
Implementing Form Views 126
User-Defined Form Views 127
Database View Classes 131
Implementing Control Views 132
Introduction to Control Views 133
Overview of CListView 135
Implementing CListView 136
Overview of CTreeView 139
Implementing CTreeView 141
Overview of CEditView 146
Implementing CEditView 147
Overview of CRichEditView 148
Implementing CRichEditView 149
Creating an Explorer-Style Application 151
Coordinating Multiple Interrelated Views 153
Sample Applications 154
Self-Check Questions 155
Chapter 5: Using Controls 157
Windows Common Controls 158
MFC Classes for Windows Common Controls 159
Creating Windows Common Controls 161
Setting Windows Common Control Attributes 164
Processing Notification Messages from Windows Common Controls 165
Internet Explorer 4.0 Common Controls 166
MFC Classes for Internet Explorer 4.0 Controls 167
Creating Internet Explorer 4.0 Controls 170
Setting Internet Explorer 4.0 Control Attributes 171
Processing Notification Messages from Internet Explorer 4.0 Common Controls 172
ActiveX Controls 173
ActiveX Controls vs. Custom Controls 174
Using Automation 175
ActiveX Control Containers 176
ActiveX Control and Container Communication 177
Implementing an ActiveX Control Container 178

Using an ActiveX Control in an MFC-based Application 179
Using ActiveX Controls on the Internet 184
Controls Supplied by MFC 185
Adding a Bitmap Button to an Application 186
Processing Notification Messages from Controls Supplied by MFC 188
Sample Applications 189
Self-Check Questions 190
Chapter 6: Creating ActiveX Controls 191
Overview of ActiveX Controls 192
Advantages of ActiveX Controls 193
Using the MFC ActiveX ControlWizard 194
Registering an ActiveX Control 198
Painting an ActiveX Control 199
ActiveX Control Properties 201
Stock Properties 202
Custom Properties 206
Using Get/Set Methods to Implement Control Properties 208
Parameterized Properties 209
Accessing Ambient Properties 210
ActiveX Control Methods 212
Adding Stock Methods 213
Adding Custom Methods 215
ActiveX Control Events 216
Adding Stock Events 217
Adding Custom Events 218
Implementing ActiveX Control Property Pages 219
Introduction to ActiveX Control Property Pages 220
Implementing the Default Property Page 221
Exchanging Data Between Controls and Property Pages 222
Adding Stock Property Pages 224
Adding Custom Property Pages 226
Creating an Enumerated Property 229
How Enumerated Properties Work 230
Implementing an Enumerated Property 231
Data Binding in an ActiveX Control 234
Introduction to Data Binding 235
How Data Binding Works 236
Creating a Bindable Property 237
Optimizing ActiveX Controls 239
Advanced Features for ActiveX Controls 240
Optimizing Drawing Code 241
Windowless Activation 242
Debugging and Handling Errors in ActiveX Applications 243
Communicating Errors in ActiveX Controls 244
Debugging ActiveX Applications 245
Sample Applications 246
Self-Check Questions 247
Chapter 7: Using OLE DB Templates for Data Access 249
Overview of OLE DB 250
Benefits of OLE DB 251
Types of OLE DB Applications 252
OLE DB Components 253
Introduction to OLE DB Templates 256
OLE DB Provider Templates 257
OLE DB Consumer Templates 259

Using CAccessor to Create a Consumer Application 260
Creating a Base Application Using MFC AppWizard 261
MFC AppWizard Implementation of CAccessor 266
COleDBRecordView Class 269
Performing Database Operations 271
Using CDynamicAccessor to Create a Consumer Application 274
Using CManualAccessor to Create a Consumer Application 277
Sample Applications 286
Self-Check Questions 287
Chapter 8: Creating ADO Database Applications 289
Introduction to ADO 290
Benefits of Using ADO 291
ADO Object Model 292
Using ADO in MFC-based Applications 294
Using Data Controls 296
The ADO Data Control 297
Data-bound Controls 299
Implementing ADO Using Data Controls 300
Using the Data-bound Dialog Box 306
Building an Application Using the Wizard 307
Default Implementation of Data-bound Dialog Box 319
Building an Application Without the Wizard 323
Adding Records 331
Editing Records 333
Deleting Records 335
Performing Queries 336
Performing a Simple Query 337
Using Filters 338
Searching for Records 339
Sample Applications 340
Self-Check Questions 341
Chapter 9: Building Internet Applications 343
Basic Internet Concepts 344
The Internet Framework 345
Types of Internet Applications 346
MFC Support for Internet Applications 347
Using the Internet Explorer Object 349
Properties and Methods 350
Controlling Internet Explorer 351
Using Automation 352
Using the Web Browser Control 354
Properties, Methods, and Events 355
Adding the Web Browser Control 356
Creating the Web Browser Control 359
Implementing the Web Browser Control 361
Implementing HTML Views 365
Using the WinInet Classes 367
Connecting to the Internet 368
Writing HTTP Applications 372
Writing FTP Applications 375
Writing Gopher Applications 377
Using the WinSock Classes 378
Introduction to Sockets 379
Using TCP/IP for Socket Communication 381
The WinSock API 382
MFC Support for WinSock 384

Sample Applications....385
Self-Check Questions....386
Chapter 10: Printing and Print Preview....387
Adding Default Printer Support....388
Introduction to the Printing Process....389
Enabling Default Printer Support....390
MFC Support for Printing....391
Screen Display vs. Printing....393
Enhancing Printer Support....394
Determining Printer Characteristics....395
Retrieving Information About the Current Print Job....396
Retrieving Information About the Printer Device Context....398
Overriding CView-Derived Functions....399
Sample Applications....405
Self-Check Questions....406